AST2000 Part 7D: Exploring Our Star

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Solar system seed 42716. Our star has mass $M = 1.187\,M_\odot$, radius $R = 1.033\,R_\odot$ $(718,735\,\mathrm{km})$ and surface temperature $T = 5882\,\mathrm{K}$. The numbers below are computed by `Project 7/star.py`.

1. Is our star a main-sequence star?

A black body of radius R and surface temperature T radiates σT^4 per unit area from its whole surface:

$$L = 4\pi R^2 \sigma T^4 = 4.406 \times 10^{26}\,\mathrm{W} = 1.15\,L_\odot. \tag{1}$$

In an HR diagram (Fig. 1) our star sits on the main sequence, right next to the Sun. That fits its mass: $1.187\,M_\odot$ is slightly more than the Sun, so on the main sequence it should be slightly hotter and brighter, as it is. It is a main-sequence star, burning hydrogen in its core.

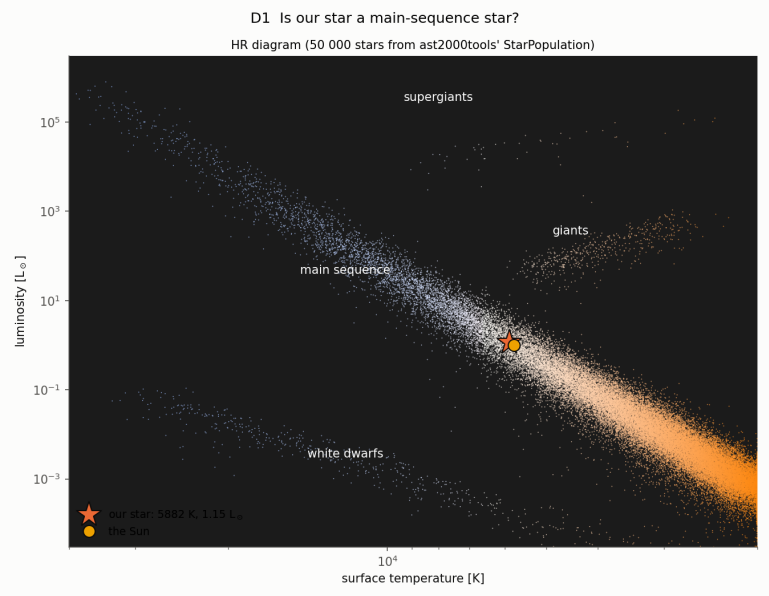


Figure 1: HR diagram: 50 000 stars from `ast2000tools`' `StarPopulation`, our star and the Sun.

2. Core temperature

Assumptions:

- uniform density $\rho_0 = M/(\frac{4}{3}\pi R^3) = 1517\,\mathrm{kg/m^3}$;
- hydrostatic equilibrium;

- an ideal gas;
- composition 75% hydrogen and 25% helium by mass.

(a) The mass profile

With uniform density, the mass inside radius r is the density times the volume of that sphere:

$$M(r) = \int_0^r \rho_0 \, 4\pi r'^2 \, \mathrm{d}r' = \frac{4\pi}{3}\rho_0 r^3 = M\frac{r^3}{R^3}. \tag{2}$$

(b) The temperature gradient

Hydrostatic equilibrium. A shell of gas does not move: the pressure difference across it balances the gravity of the mass inside it (only the mass inside pulls, by the shell theorem):

$$\frac{\mathrm{d}P}{\mathrm{d}r} = -\frac{GM(r)\rho_0}{r^2} = -\frac{4\pi}{3}G\rho_0^2 \, r. \tag{3}$$

Ideal gas. $P = nkT = \rho_0 kT/(\mu m_H)$. Since ρ_0 is constant, $\mathrm{d}P/\mathrm{d}r = (\rho_0 k/\mu m_H)\,\mathrm{d}T/\mathrm{d}r$. Equating the two expressions gives

$$\frac{\mathrm{d}T}{\mathrm{d}r} = -\frac{4\pi}{3}G\rho_0\frac{\mu m_H}{k} \, r. \tag{4}$$

Mean molecular weight. In the hot core all atoms are fully ionised. Per unit mass there are X/m_H hydrogen nuclei, each giving 2 particles (proton and electron), and $Y/4m_H$ helium nuclei, each giving 3 particles (nucleus and 2 electrons). So $1/\mu = 2X + \frac{3}{4}Y$, which gives $\mu = 0.593$.

(c) The core temperature

Integrating from the centre to the surface:

$$T(R) - T_c = -\frac{4\pi}{3}G\rho_0\frac{\mu m_H}{k}\int_0^R r \, \mathrm{d}r = -\frac{2\pi}{3}G\rho_0\frac{\mu m_H}{k}R^2, \tag{5}$$

so

$$T_c = T(R) + \frac{2\pi}{3}G\rho_0\frac{\mu m_H}{k}R^2 = 7.87\times10^6\,\mathrm{K}. \tag{6}$$

The temperature falls as a parabola from the centre to the surface (Fig. 2, left).

If the gas were *neutral* ($1/\mu = X + Y/4$, $\mu = 1.23$), T_c would double to 16.3 MK, close to the real Sun's 15.7 MK. That agreement is accidental: the uniform-density model underestimates the core temperature, and the wrong μ happens to compensate.

3. Energy production and luminosity

Since $T_c = 7.87$ MK is far below 90 MK, the core fuses hydrogen through the pp chain and the CNO cycle (pp + CNO). The core composition is $X = 0.745$, $Y = 0.253$, $X_{\mathrm{CNO}} = 0.002$. Following the AST2000 lecture notes, the energy released per kg of core is approximated by power laws in temperature ($T_6 = T/10^6\,\mathrm{K}$):

$$\varepsilon_{pp} = \varepsilon_{0,pp}X^2\rho T_6^4, \qquad \varepsilon_{0,pp} = 1.08\times10^{-12}\,\mathrm{W\,m^3\,kg^{-2}}, \tag{7}$$

$$\varepsilon_{\mathrm{CNO}} = \varepsilon_{0,\mathrm{CNO}}X X_{\mathrm{CNO}}\rho T_6^{20}, \qquad \varepsilon_{0,\mathrm{CNO}} = 8.24\times10^{-31}\,\mathrm{W\,m^3\,kg^{-2}}. \tag{8}$$

Reactions happen in the core, $r < 0.2R$, which holds a fraction $0.2^3 = 0.008$ of the mass. Its temperature is taken as T_c throughout.

- $\varepsilon_{pp} = 3.49 \times 10^{-6}\,\mathrm{W/kg}$.
- $\varepsilon_{\mathrm{CNO}} = 1.55 \times 10^{-12}\,\mathrm{W/kg}$, negligible at this temperature (its T^{20} only takes over above $\sim$18 MK).

The luminosity is then

$$L_{\mathrm{core}} = (\varepsilon_{pp} + \varepsilon_{\mathrm{CNO}})\, M_{\mathrm{core}} = 6.59 \times 10^{22}\,\mathrm{W} = 1.7 \times 10^{-4}\, L_{\odot}, \tag{9}$$

which is 6687 times smaller than the luminosity from question 1. The surface temperature this luminosity implies,

$$T = \left(\frac{L_{\mathrm{core}}}{4\pi R^2 \sigma} \right)^{1/4} = 650\,\mathrm{K}, \tag{10}$$

is far below the observed 5882 K.

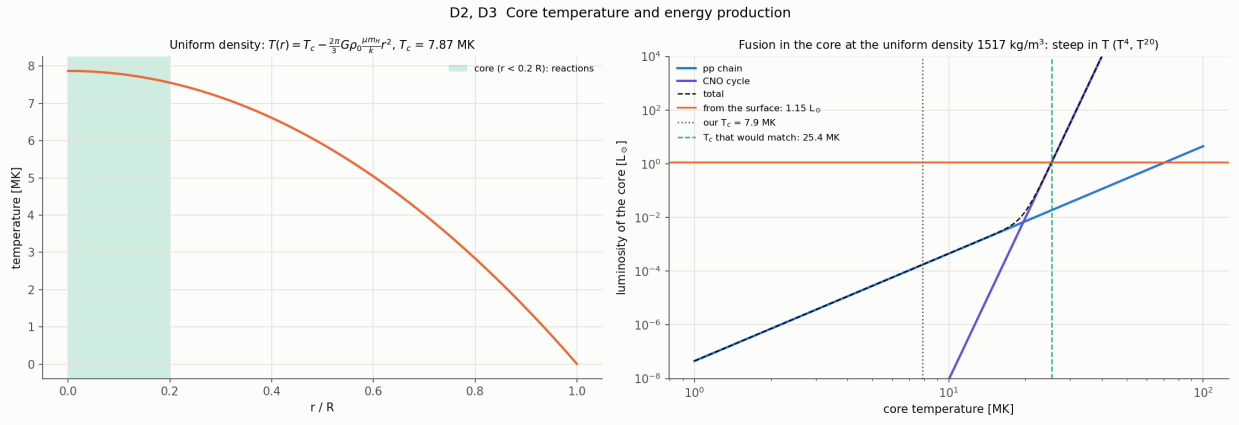


Figure 2: Left: temperature profile of the uniform-density star. Right: luminosity of the core against core temperature, for the pp chain and the CNO cycle, compared with the observed luminosity.

Discussion

The model fails by almost four orders of magnitude. The culprit is the uniform-density assumption, and Fig. 2 (right) shows why it matters so much.

- **Density.** A real star is centrally condensed. The Sun's centre is about 100 times denser than its average. A uniform star underestimates the core density, and $\varepsilon \propto \rho$.
- **Temperature.** For the same reason it underestimates the core pressure, and so the temperature. With $\varepsilon \propto T^4$ (pp) and T^{20} (CNO), small temperature errors become huge luminosity errors. Matching the observed luminosity at our ρ_0 would need $T_c \approx 25.4$ MK; with the higher real density, less would do.
- **Constant core temperature.** Assuming one temperature throughout the core overestimates the outer parts of the core, but this is a smaller effect.

Question 1, which uses only the observed radius and temperature, is the reliable value. The core model shows that the observed luminosity requires a hot, dense, centrally condensed core.